

▼ **Figure 20.14** Jennifer Doudna holding a model of CRISPR-Cas9.



➔ **Mastering Biology** HHMI Video: Jennifer Doudna: CRISPR-Cas Proteins are the Swiss Army Knife of the Genome



for example, they cannot transmit disease. An extra twist to this approach is to engineer the new allele so that it is much more highly favored for inheritance than the wild-type allele. This technology is called a **gene drive** because the biased inheritance of the engineered gene during reproduction rapidly “drives” the new allele through the population.

Other Methods for Studying Gene Function

Another method for silencing expression of selected genes doesn’t alter the genome; instead, it exploits the phenomenon of **RNA interference (RNAi)**, described in Concept 18.3. This experimental approach uses synthetic double-stranded RNA molecules matching the sequence of a particular gene to trigger breakdown of the gene’s messenger RNA or to block its translation. In organisms such as the nematode and the fruit fly, RNAi has already proved valuable for analyzing the functions of genes on a large scale. This method is quicker than using the CRISPR-Cas9 system, but it leads to only a temporary reduction of gene expression rather than a permanent gene knockout or alteration.

In humans, ethical considerations prohibit knocking out genes to determine their functions. An alternative approach is to analyze the genomes of large numbers of people with a certain phenotypic condition or disease, such as heart disease or diabetes, to try to find differences they all share compared with people without that condition. The logic is that these differences may be associated with one or more malfunctioning genes and thus in a sense are naturally occurring gene knockouts. In these large-scale analyses, called **genome-wide association studies**, researchers look for *genetic markers*, DNA sequences that vary in the population. In a gene, such sequence variation is the basis of different alleles,

as we have seen for sickle-cell disease (see Figure 17.26). And just like the coding sequences of genes, noncoding DNA at a specific locus on a chromosome may exhibit small nucleotide differences among individuals. Variations in coding or noncoding DNA sequences among a population are called **polymorphisms** (from the Greek for “many forms”).

➔ **Mastering Biology** Interview with Charles Rotimi: Using genomics to study health-related conditions in African-Americans



Among the most useful genetic markers in tracking down genes that contribute to diseases and disorders are single base-pair variations in the genomes of the human population. For more than 99% of the nucleotides in the human genome, virtually all people have the same nucleotide in each position. However, once in every 100–300 base pairs of both coding and noncoding DNA sequences are positions where the sequence varies between individuals. A single base-pair site where variation is found in at least 1% of the population is called a **single nucleotide polymorphism (SNP)**, pronounced “snip”); a few million SNPs occur in the human genome. To find SNPs in large numbers of people, it isn’t necessary to sequence their DNA; SNPs can be detected by very sensitive microarray assays, RNA-seq, or PCR.

Once a SNP is identified that is found in all people affected by the disease being studied, researchers focus on that region and sequence it. In nearly all cases, the SNP itself does not contribute directly to the disease in question by altering the encoded protein; in fact, most SNPs are in noncoding regions. Instead, having a particular SNP associated with a disease suggests that the gene whose mutation causes the disease is located very close to that SNP on that chromosome. This closeness (*genetic linkage*; see Concept 15.3) means that crossing over between the SNP and the gene is very unlikely during gamete formation. Therefore, the SNP and gene are almost always inherited together, so the SNP acts as a genetic marker for the disease-causing allele (**Figure 20.15**). SNPs have been

▼ **Figure 20.15** Single nucleotide polymorphisms (SNPs) as genetic markers for disease-associated alleles. This diagram depicts the same region of the genome from two groups of individuals, one group having a particular disease or condition with a genetic basis. Unaffected people have an AT pair at a given SNP locus, while affected people have a CG pair there. Once the allele is confirmed as being associated with the disease in question, the SNP that varies in this way can be used as a marker for the disease-associated allele.

